

Spinny × Anand K

OEM Spare Parts Crawler — Automation Requirement Document

AUTOMATION REQUIREMENT DOCUMENT

OEM Spare Parts Catalogue Crawler

Field	Value
Prepared For	Anand K (Vendor)
Prepared By	Spinny
Process Owner	Spinny Catalog / OES Datasources Team
Project	Automated OEM Spare Parts Catalogue Crawler
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Companion Artefact	Spinny Catalog OES Datasources sheet (shared separately) — single source of truth for site URLs, navigation steps and per-site data parameters.

1. Objective

Automate the monthly extraction of publicly available passenger-vehicle spare-parts catalogue data from the 19 OEM and aftermarket-brand websites listed in Section 3. The crawler must replicate the human navigation workflow currently performed by the Spinny Catalog team and produce a clean, deduplicated, structured dataset for every in-scope brand once a month.

2. Purpose & Background

Spinny is India's used-car platform. Accurate, up-to-date spare-parts data — names, codes, MRPs and vehicle compatibility — is critical for downstream workstreams: refurbishment cost estimation, service-quote generation, dealer settlement, and parity checks against in-house catalogue and pricing. Today this data is collected manually by browsing each brand's website tab-by-tab, which is slow, error-prone, and cannot keep pace with how often OEMs revise their catalogues.

Automating the crawl converts a multi-day manual task into a deterministic, auditable monthly job. It gives the Pricing and Catalog teams a single consolidated dataset they can rely on, and frees catalog operations capacity for higher-value work.

3. Scope

3.1 In-Scope Brands

The following 19 brand websites are in scope for Phase 1. Every brand has a dedicated tab in the companion sheet that defines the navigation steps and the exact data parameters to extract for that site. The sheet is the single source of truth for both.

#	Brand	Website
1	HELLA	https://shop4hella.com/listing/4-wheeler
2	UNO MINDA	https://unomindakart.com/
3	Technix	https://technixauto.com/
4	GABRIEL	https://www.anandgroupindia.com/gabrielindia/aftermarkets/?pcatid=all&vcid=passenger-vehicle
5	ZIP	https://www.zipfilters.com/
6	MONROE	https://www.bksmotors.com/brand/monroe
7	AMARON	https://www.amaron.com/passenger-vehicles
8	SF SONIC	https://www.sfbatteries.in/battery-finder/4w-battery/
9	EXIDE	https://www.exideindustries.com/products/automotive-batteries/four-wheeler-batteries.aspx
10	Spark Minda	https://mcl-aftermarket.com/Product/ProductServices
11	LuK / Schaeffler	https://vehiclelifetimesolutions.schaeffler.in/en-gb/catalog
12	AUTOKOI	https://www.autokoi.com/products/
13	Bosch	https://ap.boschaftermarket.com/in/en/parts/
14	VALEO	https://www.valeoservice.in/en-in/passenger-car
15	ZF	https://aftermarket.zf.com/en/aftermarket-portal/our-catalog/search-by-vehicle/
16	LUMAX	https://www.lumaxworld.in/aftermarket/product-catalogue.html
17	JK TYRE	https://www.jktyre.com/
18	TVS Girling	https://partscatalogue.brakesindia.com/ & https://www.tvsgirling.com/passenger-cars-scv/
19	Mobil	https://www.mobil.co.in/en-in/our-products

Note: TVS Girling has two distinct URLs and two distinct navigation workflows defined in the sheet. It is treated as a single brand with two crawl targets; both are in scope and both feed into the same TVS Girling output file.

3.2 Vehicle Scope

Passenger vehicles only. Two-wheeler, three-wheeler, commercial-vehicle and tractor parts are explicitly out of scope. Where a brand surfaces a vehicle-segment filter, the crawler must select the passenger-car / 4-wheeler segment before extracting data.

4. Data Fields

Core data fields vary by brand. The exact list of fields to extract for each brand is defined in the "Relevant Data Parameters" column of that brand's tab in the companion sheet. The vendor must extract precisely those fields for each brand — no more, no less.

Common field labels that appear across brands include Item Name, Item Code, MRP, Compatible vehicle / car model, Tyre sizes, and Vehicle Information (Compatibility). The vendor will normalise these to a consistent output schema; the mapping from sheet labels to schema column names will be agreed during kick-off.

In addition to the per-site data fields, every output row will carry the following crawler-written metadata, which the vendor must populate automatically:

Metadata Field	Type	Description
source_website	String	Brand identifier and originating URL the row was scraped from.
crawl_date	Date (ISO-8601)	UTC date of the run that produced the row.
crawl_status	Enum	success partial failed — populated by the crawler runtime per row.

5. Output Format & Delivery

The crawler will produce the following artefacts per monthly run, delivered to a shared Spinny-managed cloud folder (destination to be confirmed at kick-off) in a date-stamped sub-directory.

Artefact	Format	Naming Convention	Notes
Per-brand file	CSV + JSON	<brand>_<YYYYMMDD>.csv / .json	One file per in-scope brand, containing all rows scraped that month. TVS Girling produces a single combined file across both URLs, with the source URL captured in source_website.
Consolidated master	CSV + JSON	spinny_oem_master_<YYYYMMDD>.csv / .json	Concatenated dataset across all 19 brands, deduplicated within each

Artefact	Format	Naming Convention	Notes
			brand on (source_website, item_code) where item_code exists, and on (source_website, item_name, compatible_models) otherwise.
Run summary	JSON	run_summary_<YYYYMMDD>.json	Per-brand row counts, crawl_status distribution, error count, runtime, and a list of any sites whose DOM appeared to have changed.

Both CSV and JSON outputs must use UTF-8 encoding. Numeric fields (MRP) must be unquoted in CSV and typed as numbers in JSON; the crawler must strip any currency symbols, commas and whitespace before writing.

6. Edge Cases & Crawler Design Considerations

The vendor must explicitly design for and test the following navigation patterns observed across the 19 brand workflows. Detailed step-by-step navigation for each brand is in the companion sheet; the items below are crawler-architecture observations the vendor should plan for.

Pattern	Brands exhibiting this pattern	Required crawler behaviour
Cascading dropdowns (Make → Model → Variant / Fuel)	AMARON, SF SONIC, LuK / Schaeffler, JK TYRE, ZF, TVS Girling (URL 1)	Iterate the full Cartesian product of all dropdown levels respecting upstream→downstream dependencies (downstream options reset when an upstream changes). Persist a checkpoint of completed combinations so a run can resume after interruption.
Multi-level category iteration	HELLA, UNO MINDA, ZIP, EXIDE, Spark Minda, VALEO, LUMAX, GABRIEL	Walk the category tree to its leaves; iterate every (category, sub-category) pair; follow pagination to the last page within each leaf.
Multi-URL same-brand	TVS Girling (two distinct URLs and workflows)	Run both workflows; combine into one brand output file; preserve the originating URL in source_website on every row.
Geography / region gate	ZF (must select India + Passenger cars and LCV before search)	Apply the geography / vehicle-type gate as the first step; reject results that bypass it.

Pattern	Brands exhibiting this pattern	Required crawler behaviour
Vehicle-segment filter required	GABRIEL, Spark Minda, JK TYRE, ZF	Apply the 4W / passenger-vehicle filter; reject any row whose vehicle segment is not passenger vehicle.
Catalogue lives in downloadable brochures, not HTML listings	Bosch	Open each product category, fetch the linked brochure (PDF), and parse the brochure for the required fields. Treat broken or missing brochures as <code>crawl_status = failed</code> for that category.
Hidden / off-canvas navigation	AUTOKOI (e-catalogue accessed via top-right hamburger menu)	Trigger the hidden navigation programmatically rather than relying on the default home-page DOM.
Flat product list (with potential pagination)	MONROE, Technix, Mobil	Iterate the full list; detect and follow pagination if present; verify completeness by comparing extracted count against any total-count indicator on the page.

7. Assumptions

- All 19 in-scope brand websites are publicly accessible without authentication. No login or paid subscription is required.
- No CAPTCHA, hCaptcha, reCAPTCHA, or equivalent challenge is presented at the point of data extraction on any in-scope site.
- MRP, where extracted, is in INR. The crawler will not perform any FX conversion.
- Only passenger-vehicle (4W) parts are in scope; commercial vehicles, two-wheelers, three-wheelers and tractors are excluded.
- The 'Relevant Data Parameters' column in each brand tab of the companion sheet is the authoritative list of fields the crawler must extract for that brand. Where a field listed in the sheet is not visible on the live site for a given product, that field is left blank for that row and `crawl_status` is set to `partial`.
- Site DOM, URL structure and filter behaviour are reasonably stable between monthly runs; structural changes will be flagged via the SLA process in Section 8.
- `robots.txt` and the site's terms of use permit polite, low-frequency crawling at the cadence and volume defined in this document.
- The vendor is responsible for crawl politeness — request rate limiting, user-agent disclosure, and back-off on 4xx / 5xx responses.

8. SLAs & Change Handling

Item	Commitment
Crawl frequency	Monthly. Run window: the night of the first Monday of each calendar month, between 22:00 IST and 06:00 IST (Tuesday).

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Output delivery	Output files must land in the shared folder no later than 12:00 IST on the Tuesday following the run.
Per-run completeness target	≥ 98% of in-scope brands return crawl_status = success per monthly run.
Site breakage detection	Vendor must monitor each monthly run and detect site breakages (DOM change, filter change, blocked response) automatically.
Breakage notification SLA	Vendor must notify Spinny within 24 hours of detecting a broken site.
Breakage resolution SLA	Vendor must restore a broken site to crawl_status = success within 5 business days of detection.
Out-of-cycle change requests	Adding a new site, changing data fields, or changing crawl frequency follows a change-request flow with mutually agreed effort estimate and timeline.
Data retention	All monthly outputs retained in the shared folder for at least 12 months.